

Secret Sharing Scheme

Theoretical Overview

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Why Secret Sharing?

- Allows splitting a secret into *shares* so that only certain subsets of participants Γ can reconstruct the secret information.
- Avoid single points of failure: Attribute-Based Encryption (**ABE**) - key management, authority, or storage.
- Foundational block for many cryptographic protocols: secure multi-party computation, threshold cryptography, access control, multi-factor authentication.

Motivation II

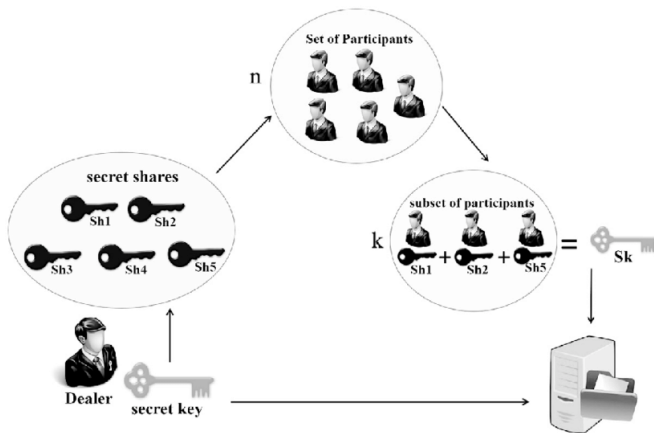


Figure: The notion of Threshold Secret Sharing [GAQ20]

- Parties: $\mathcal{P} = \{P_1, \dots, P_n\}$.
- Access structure $\mathcal{A} \subseteq 2^{\mathcal{P}}$: sets that can reconstruct.
- **Monotone** requirement: if $A \in \mathcal{A}$ and $A \subseteq B$, then $B \in \mathcal{A}$.

Perfect Secret Sharing - [Bei11]

Let secret S be random and shares be random variables $\{S_i\}$. The domain of secrets S (given by its shares) is a secret sharing scheme realizing an access structure $\mathcal{A}$ if:

- **Correctness:** For every $A \in \mathcal{A}$, S is determined by $\{S_i : P_i \in A\}$.
- **Perfect privacy:** For every $T \notin \mathcal{A}$, shares reveal no information:

$$I(S; \{S_i : P_i \in T\}) = 0 \iff H(S \mid \{S_i : i \in T\}) = H(S).$$

- Lower bound intuition: in any perfect scheme, each non-redundant party's share must be at least as large as the secret.

Threshold Secret Sharing I

Shamir's t, n Threshold Scheme

- Is based on **threshold** access structure t -out-of- n :

$$\mathcal{A}_t = \{A \subseteq \mathcal{P} : |A| \geq k\}.$$

- Work over a finite field $\mathbb{F}_q$ with $q > n$
- Choose a random polynomial of degree $t - 1$:

$$P(x) = s + \alpha_1 x + \cdots + \alpha_{t-1} x^{t-1}, \quad \alpha_i \in \mathbb{F}_q.$$

- Give share $P(i)$ to party P_i .
- Any t shares interpolate $P(x)$ and recover $s = P(0)$; any $t - 1$ shares reveal nothing.

Why Shamir is a Baseline

- Perfect privacy and correctness for threshold structures.

Threshold Secret Sharing II

- Ideal information rate for threshold access structures [Bri89].
- Linearity enables efficient protocols and MPC constructions.
- Limitation: primarily threshold policies unless generalized.

The need for a General Access Structure

- Real policies are not always “any t out of n ”.
- General secret sharing schemes exist for any monotone access structure, but efficiency is the main challenge [Bei11].

Benaloh-Leichter Approach

- Represent an access policy as a monotone formula over parties.
- If an access structure can be described by a *small monotone formula* → efficient perfect secret sharing-scheme.
- For $F = F_1 \vee \dots \vee F_k$: replicate the secret down each branch.
- For $F = F_1 \wedge \dots \wedge F_k$: split the secret into k parts using a (k, k) scheme, then recurse.
- Formula size directly affects share size blow-up.

Efficiency Challenge for General Access Structures

- General constructions can have exponential share size in the worst case.
- There is a significant gap between known upper bounds and lower bounds for some families.
- Lower bounds and structural results for linear and multilinear schemes are surveyed in.

Ideal Secret Sharing and Structural Characterizations - [Bri89]

- Ideal scheme: share size equals secret size.
- Threshold structures are ideal; some non-threshold structures are also ideal.
- Links between ideality and combinatorial structures (e.g., matroids) appear in the classical literature.

Thank you!

Questions?



Amos Beimel.

Secret-sharing schemes: A survey.

In *International conference on coding and cryptology*, pages 11–46.
Springer, 2011.



Ernest F Brickell.

Some ideal secret sharing schemes.

In *Workshop on the Theory and Application of Cryptographic Techniques*, pages 468–475. Springer, 1989.



Adnan Gutub and Adel Al-Qurashi.

Secure shares generation via m-blocks partitioning for counting-based secret sharing.

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